Math review: number skills

Use this page when signs, decimals, or fractions get in the way of the science.

1 Positive and negative numbers

On a number line, positive is right of zero and negative is left. In physics, a sign can show direction: for example, east + and west -.

Operation	How to do it	Example
Add different signs	Subtract sizes; keep the sign of the number farther from zero.	14 + (-9) = 5 -14 + 9 = -5
Subtract	Add the opposite of the second number.	4 - 11 = 4 + (-11) = -7 4 - (-3) = 4 + 3 = 7
Multiply or divide	Same signs → positive. Different signs → negative.	(-12) ÷ 3 = -4 (-3) × (-2) = 6

Try: A. 9 + (-14) = _____ B. 2 - (-5) = _____ C. -18 ÷ 6 = _____

2 Decimals

Add/subtract: line up decimal points. Add zeros when helpful: 6.00 - 2.75 = **3.25**.
Multiply: multiply whole numbers first, then place the decimal. For 1.2 × 0.4, do 12 × 4 = 48; two decimal places gives **0.48**.
Divide: shift both decimal points the same amount so the divisor is a whole number: 7.2 ÷ 0.6 = 72 ÷ 6 = **12**.

Estimate first: 9.6 ÷ 3 should be a little more than 3, not 32 or 0.32.
Try: D. 5.4 - 1.85 = _____ E. 0.3 × 8 = _____ F. 4.5 ÷ 0.5 = _____

3 Fractions are division

Meaning: 3/4 means 3 ÷ 4 = **0.75**. A fraction bar groups the whole numerator: (12 - 4)/2 = 8/2 = **4**.
Multiply: multiply tops and bottoms: (2/3) × (3/4) = 6/12 = **1/2**.
Divide: multiply by the reciprocal (flip the second fraction): (3/4) ÷ (1/2) = (3/4) × (2/1) = 6/4 = **1.5**.
Add/subtract: use the same denominator first: 1/2 + 1/4 = 2/4 + 1/4 = **3/4**.

Try: G. 3/5 = _____ (decimal) H. (9 - 3)/2 = _____
I. (1/2) × (2/3) = _____ J. (2/3) ÷ (1/3) = _____

Check your practice
A -5; B 7; C -3; D 3.55; E 2.4; F 9; G 0.6; H 3; I 1/3; J 2.

Math review: equations & units

A letter stands for a number. Undo operations to find its value.

4 One-step equations: keep both sides equal

Equation	Undo it on BOTH sides	Solution
x + 7 = 19	Subtract 7: x = 19 - 7	x = 12
x - 6 = 11	Add 6: x = 11 + 6	x = 17
4x = 18	Divide by 4: x = 18 ÷ 4	x = 4.5
x/5 = 3	Multiply by 5: x = 3 × 5	x = 15

Check by substitution: for 4x = 18, replace x with 4.5. Since 4 × 4.5 = 18, the solution works.
Try: K. x + 8 = 15 L. 6x = 21 M. x/4 = 2.5

5 Substitute carefully; follow the order

Work in this order: **parentheses**, then multiply/divide from left to right, then add/subtract from left to right.
Example: a = (vf - vi)/t, where vf = 2, vi = 8, t = 3.
1. Write a = (2 - 8)/3. **2.** Subtract inside parentheses: -6/3.
3. Divide: a = **-2**. In this physics formula, the unit is m/s².
Do not enter 2 - 8 ÷ 3; that leaves out the grouping.

Try: N. (4 - 10)/2 = _____ O. 3 + 0.5 × 8 = _____

6 Match units before you calculate

1 minute = 60 seconds • 1 kilometer = 1,000 meters • 1,000 grams = 1 kilogram.
To get m/s, divide meters by seconds. Convert a time in minutes before using it with meters: 1.5 min × 60 = **90 s**.
To use Fnet = ma, convert mass to kg: 750 g ÷ 1,000 = **0.75 kg**.
m/s² means “meters per second, each second.” It is a unit of acceleration, not a request to square the final answer.

Try: P. 2.5 min = _____ s Q. 400 g = _____ kg
R. 0.8 km = _____ m

Finish well: Keep calculator digits until the end. Round to two decimal places if needed: 10 ÷ 3 ≈ 3.33. Write the **unit** and **direction** when needed. Distance and speed cannot be negative.

Check your practice
K 7; L 3.5; M 10; N -3; O 7; P 150 s; Q 0.4 kg; R 800 m.